

Modulo Didattico 2 – Scale di misura e illustrazioni grafiche statistica univariata

Corso Analisi dei dati con applicazioni informatiche

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A.A. 2026/2027

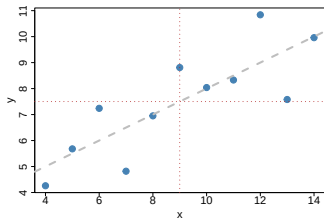
Graphs remind us that the process of statistical inference is not mechanical. This process often involves subjective decisions (e.g., the evaluation, exploration, and/or deletion of outliers) which are an integral part of the analysis. Thus, graphs are among the most appropriate tools to enhance transparency and confer plausibility to such decisions.

Pastore et al. (2017)

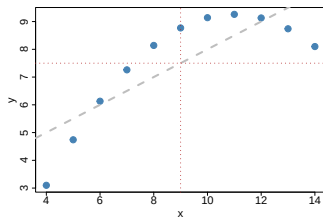
Data set di Anscombe: Quattro dataset, di seguito le statistiche descrittive relative alla variabile x e alla variabile y e la loro correlazione:

Table 1: Dataset di Anscombe

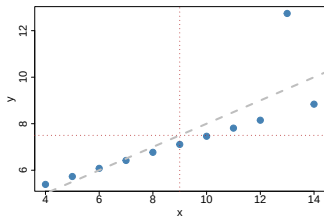
Dataset	Media.X	Media.Y	Varianza.X	Varianza.Y	Correlazione
Dataset 1	9	7.5	11	4.13	0.82
Dataset 2	9	7.5	11	4.13	0.82
Dataset 3	9	7.5	11	4.12	0.82
Dataset 4	9	7.5	11	4.12	0.82



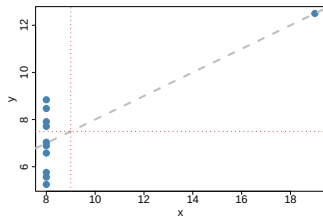
(a) Dataset 1



(b) Dataset 2



(c) Dataset 3



(d) Dataset 4

Figure 1: I grafici raccontano un'altra storia

1 Grafici

2 Esempi

R fornisce uno strumento molto utile per la rappresentazione grafica dei dati e dei risultati già con il suo pacchetto base `graphics`

NB

Non avete bisogno di installarlo, è caricato automaticamente nel momento in cui scaricate R e le sue funzioni sono sempre disponibili

`ggplot2` è un pacchetto più avanzato per la creazione dei grafici. Non entreremo nel dettaglio del suo funzionamento perché è un po' più complicato

Alcune funzioni incluse nel pacchetto `graphics`:

<code>plot(x)</code>	①
<code>plot(x, y)</code>	②
<code>hist(x)</code>	③
<code>boxplot(x)</code>	④
<code>barplot(x)</code>	⑤

- ① Grafico di una variabile quantitativa
- ② Grafico a dispersione di due variabili quantitative
- ③ Istogramma di una variabile quantitativa
- ④ Boxplot della distribuzione quartile di una (o più) variabili
- ⑤ Grafico a barre per le frequenze delle variabili qualitative

1 Grafici

- Grafico a barre
- Istogrammi
- Boxplot
- Boxplot – Calcolo degli outlier

2 Esempi

Usato per variabili qualitative nominali o ordinali:

- Asse x : categoria k
- Asse y : Frequenza n_k (vale anche per le frequenze percentuali)

L'altezza della barretta è proporzionale alla sua frequenza

`barplot(x)`

Dove x è il vettore delle frequenze

Grafico a barre

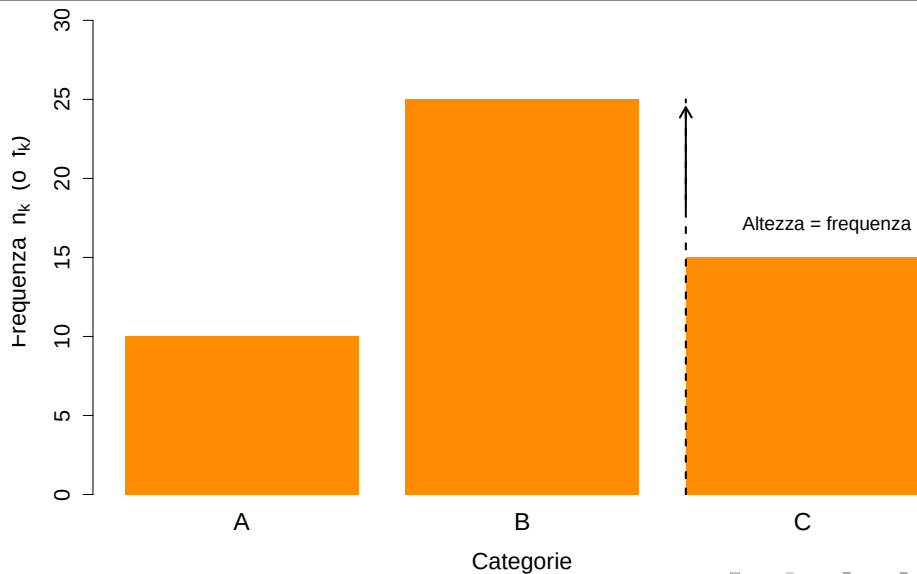


Table 2: Tabelle di frequenza per le categorie A, B, C.

Categoria	n_k
A	21
B	24
C	22

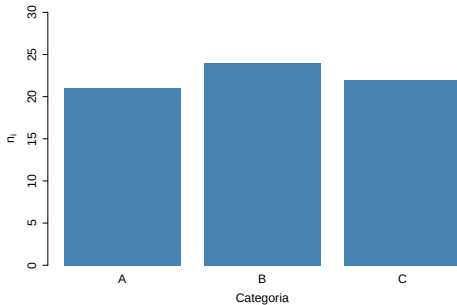


Figure 2: Barplot delle tabelle

Categoria	n_k
A	12
B	35
C	18

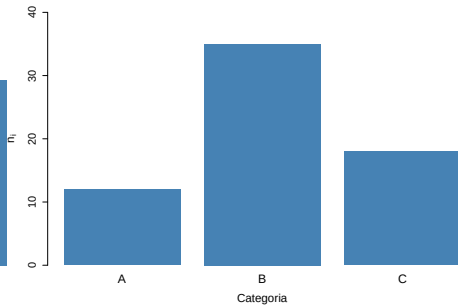


Figure 3: Barplot delle tabelle

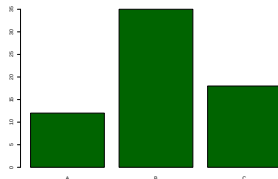
```
dati = rep(c("A", "B", "C"), c(12, 35, 18)); dati
```

[illegible]

```
table(dati)
```

```
dati
  A  B  C
12 35 18
```

```
barplot(table(dati), col = "darkgreen")
```



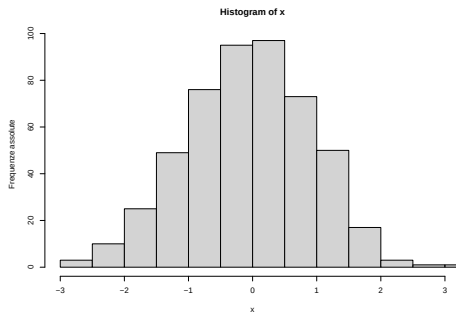
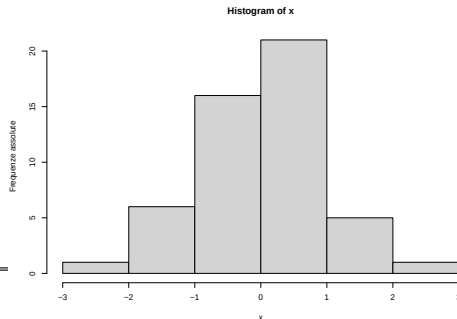
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2 Esempi

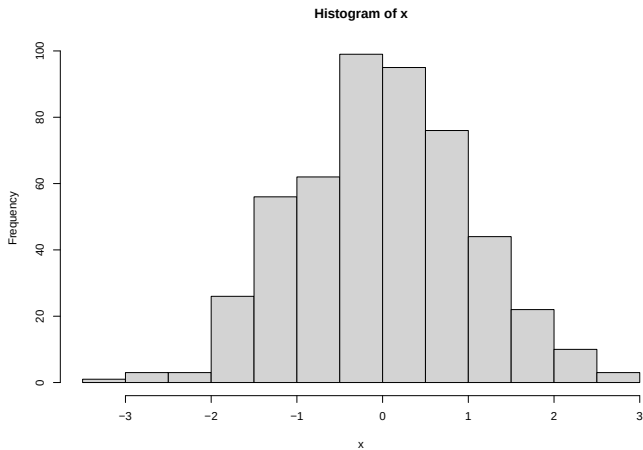
Istogrammi

`hist(x, breaks)` Produce un istogramma a partire da un vettore `x` di osservazioni quantitative. `breaks` definisce la larghezza delle singole barre, secondo diverse regole

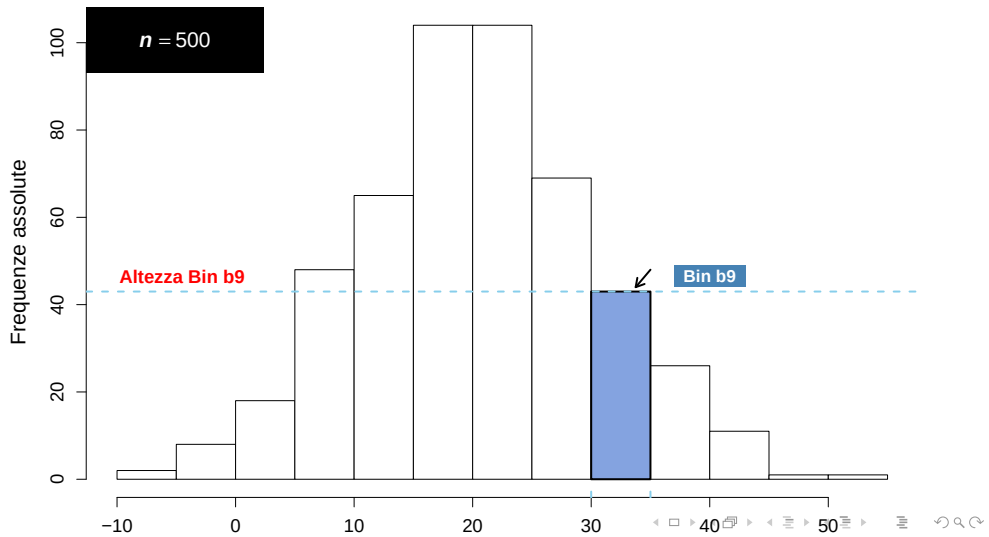
(a) $n = 500$ (a) $n = 50$

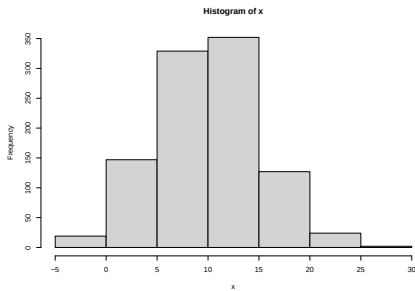
Istogrammi con n diverso

```
x = rnorm(500, mean = 0, sd = 1)
hist(x)
```

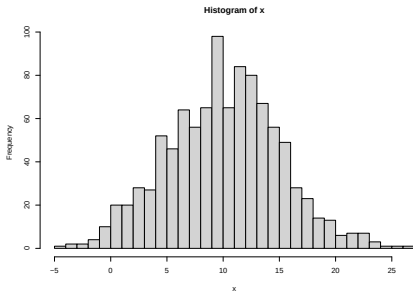


Histogram of X

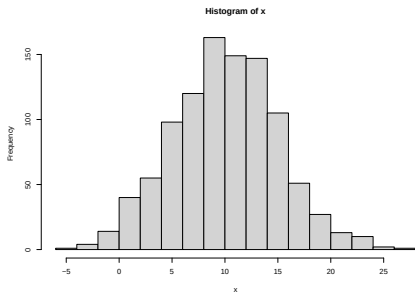




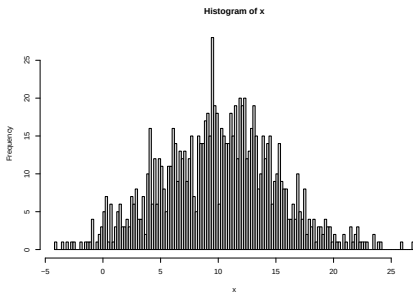
(a) Sturges - Default



(b) FD



(c) Scott



(d) User defined (145)

Sturges:

$$\frac{\max(x) - \min(x)}{\log_2(n + 1)}$$

Scott:

$$3.5 \sqrt{\text{var}(x)} n^{-1/3}$$

Freedman-Diaconis

$$2 (Q_3 - Q_1) n^{-1/3}$$

User defined:

$$\frac{\max(x) - \min(x)}{\text{breaks}}$$

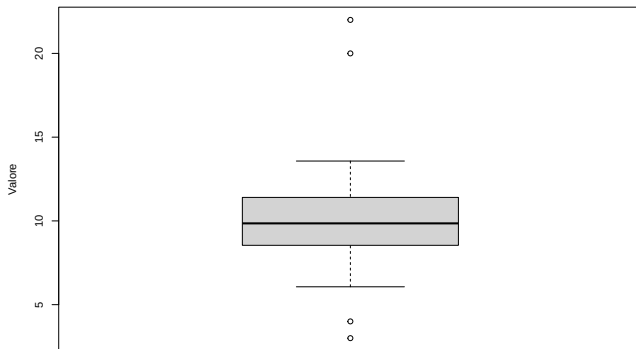
1 Grafici

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2 Esempi

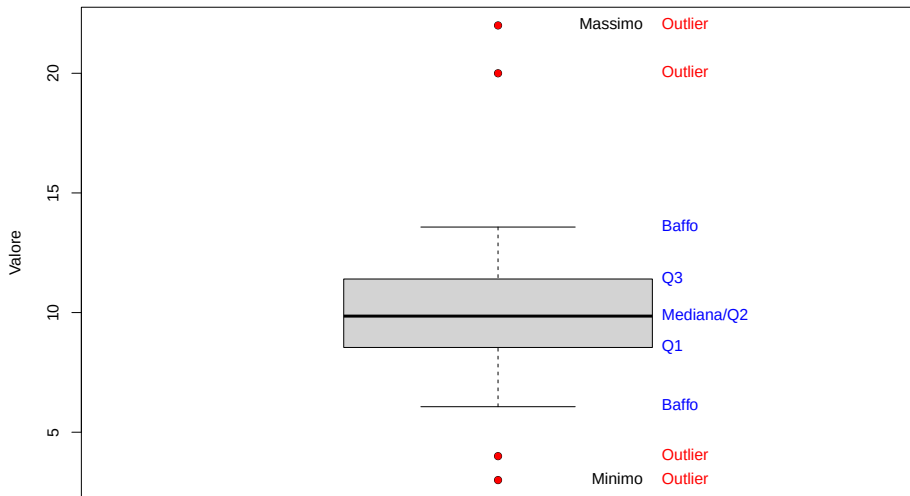
Boxplot

`boxplot(x)` Produce un boxplot a partire da un vettore X di osservazioni quantitative



Boxplot

Boxplot



1 Grafici

- Grafico a barre
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2 Esempi

La lunghezza dei baffi (e di conseguenza la definizione degli outliers) è definita attraverso il metodo Tukey:

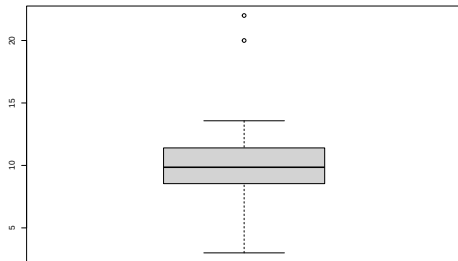
$$[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$$

dove $IQR = Q_3 - Q_1$ (Inter-quartile range)

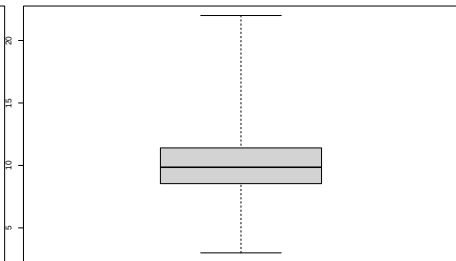
Settando l'argomento `range = x` si può cambiare il modo in cui vengono calcolati i baffi:

```
boxplot(x, range = 2)
```

```
boxplot(x, range = 0)
```



(a) `range = 2` - Meno Conservativo



(a) `range = 0` - Minimo e Massimo

Calcolo degli outlier

1 Grafici

2 Esempi

barplot()

```
voto = sample(rep(c("si", "no", "astenuti"),
                  c(12, 15, 2)))
voto = factor(voto, levels = c("si", "no", "astenuti"))
table(voto)

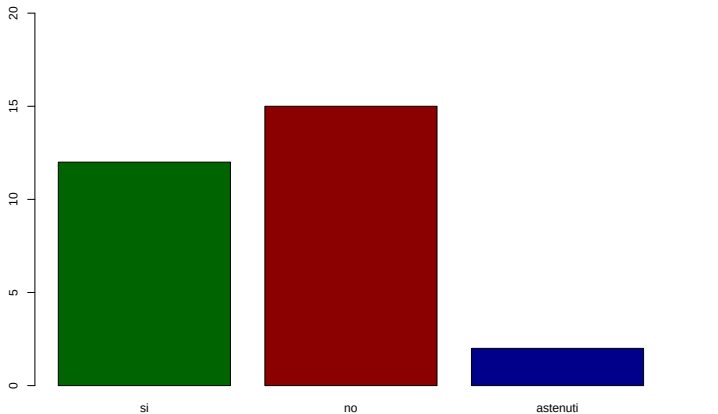
# le frequenze restituite dalla funzione table() vanno usate
# come vettore di base per il barplot

barplot(table(voto),
        col = c("darkgreen", "darkred", "darkblue"),
        ylim = c(0, 20))
```

```
voto
```

si	no	astenuti
12	15	2

```
barplot(table(voto),  
  col = c("darkgreen", "darkred", "darkblue"),  
  ylim = c(0, 20))
```



boxplot()

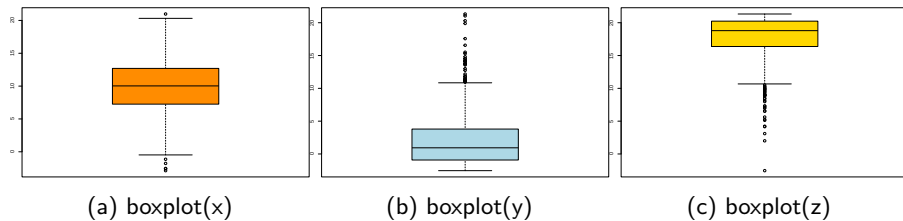
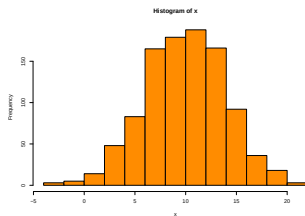
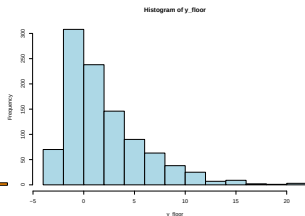


Figure 9: Distribuzioni a confronto

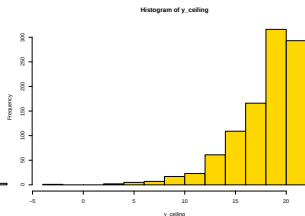
hist()



(a) hist(x)



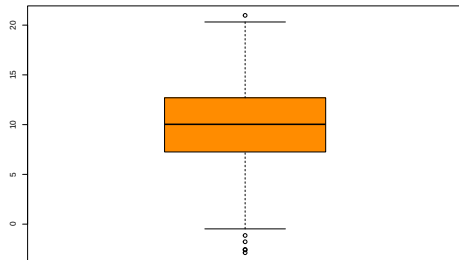
(b) hist(y)



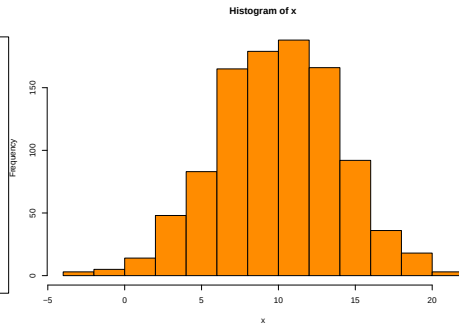
(c) hist(z)

Figure 10: Distribuzioni a confronto

Confronto boxplot vs istogramma



(a) `boxplot(x)`



(b) `hist(x)`

Figure 11: Boxplot vs Istogramma

Confronto boxplot vs istogramma

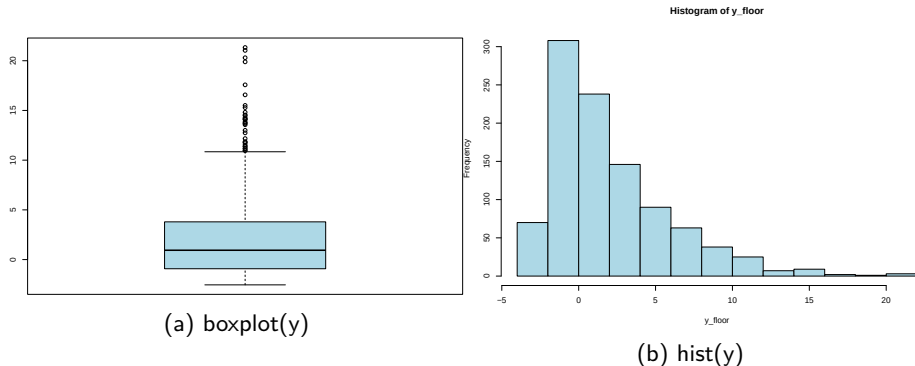


Figure 12: Boxplot vs Istogramma

Confronto boxplot vs istogramma

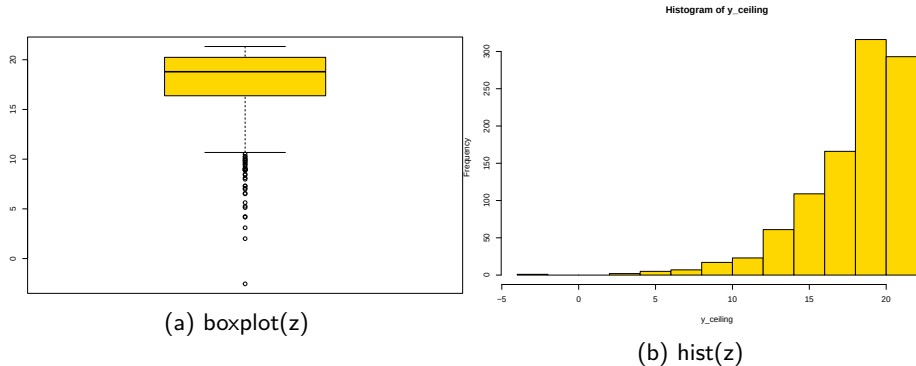
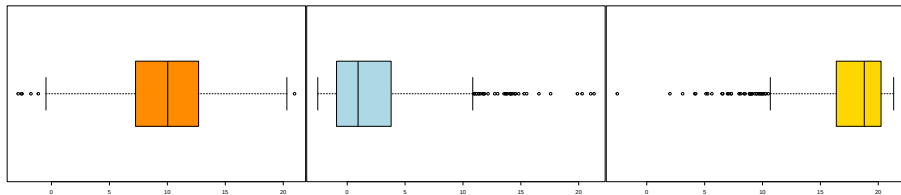


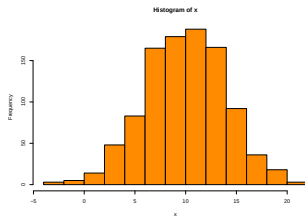
Figure 13: Boxplot vs Istogramma



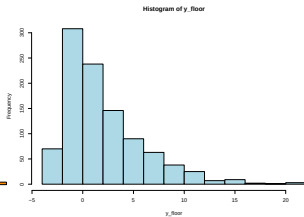
(a)

(b)

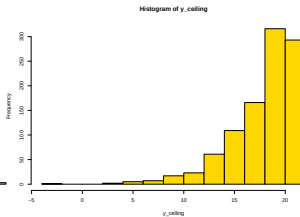
(c)



(d)



(e)



(f)